Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

TOPIC: DDL, DML, DCL, TCL COMMANDS

DATABASE: BANK MANAGEMENT SYSTEM

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Step 1 — Create Database

```sql
CREATE TABLE bank_transactions (
    txn_id INT PRIMARY KEY,
    customer_name VARCHAR(50),
    branch_name VARCHAR(50),
    transaction_type VARCHAR(20),
    amount DECIMAL(10 , 2 ),
    transaction_date DATE
);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
27	09:07:38	CREATE INDEX idx_books_isbn ON Books(isbn)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
28	09:07:38	SELECT * FROM Books WHERE isbn = '9780451524935' LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
29	09:17:32	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_n...	0 row(s) affected	0.031 sec

Step 2 ALTER TABLE ADD COLUMN:

```sql
ALTER TABLE bank_transactions
ADD account_no VARCHAR(20);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
28	09:07:38	SELECT * FROM Books WHERE isbn = '9780451524935' LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
29	09:17:32	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_n...	0 row(s) affected	0.031 sec
30	09:19:23	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec

Step 3 ALTER TABLE MODIFY COLUMN

```
11
12 •  ALTER TABLE bank_transactions
13    MODIFY customer_name VARCHAR(100);
14
15
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
29	09:17:32	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_n...	0 row(s) affected	0.031 sec
30	09:19:23	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
31	09:20:25	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec

Step 4 RENAME TABLE

sql week 1 | sql week 2 | SQL File 4*

Limit to 1000 rows

```
7         branch_name VARCHAR(50),
8         transaction_type VARCHAR(20),
9         amount DECIMAL(10, 2),
10        transaction_date DATE
11    );
12
13 •  ALTER TABLE bank_transactions
14    ADD account_no VARCHAR(20);
15
16 •  ALTER TABLE bank_transactions
17    MODIFY customer_name VARCHAR(100);
18
19 •  RENAME TABLE bank_transactions TO customer_transactions;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
11	11:23:05	CREATE DATABASE bank_db	1 row(s) affected	0.016 sec
12	11:23:05	USE bank_db	0 row(s) affected	0.000 sec
13	11:23:05	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_na...	0 row(s) affected	0.015 sec
14	11:23:05	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
15	11:23:05	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
16	11:23:05	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected	0.016 sec

STEP 5 TRUNCATE TABLE

```
21 •  TRUNCATE TABLE customer_transactions;
22
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
16	11:23:05	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected	0.016 sec
17	11:24:13	TRUNCATE TABLE bank_transactions	Error Code: 1146. Table 'bank_db.bank_transactions' doesn't exist	0.016 sec
18	11:25:36	CREATE DATABASE bank_db	Error Code: 1007. Can't create database 'bank_db'; database exists	0.000 sec
19	11:25:41	CREATE DATABASE bank_db	Error Code: 1007. Can't create database 'bank_db'; database exists	0.000 sec
20	11:25:46	CREATE DATABASE bank_db	Error Code: 1007. Can't create database 'bank_db'; database exists	0.000 sec
21	11:25:57	TRUNCATE TABLE customer_transactions	0 row(s) affected	0.015 sec

STEP 6 DROP TABLE

```
25 •  DROP TABLE bank_backup;
26
27
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
19	11:25:41	CREATE DATABASE bank_db	Error Code: 1007. Can't create database 'bank_db'; database exists	0.000 sec
20	11:25:46	CREATE DATABASE bank_db	Error Code: 1007. Can't create database 'bank_db'; database exists	0.000 sec
21	11:25:57	TRUNCATE TABLE customer_transactions	0 row(s) affected	0.015 sec
22	11:26:56	DROP TABLE bank_backup	Error Code: 1051. Unknown table 'bank_db.bank_backup'	0.000 sec
23	11:28:30	CREATE TABLE bank_backup AS SELECT * FROM customer_transactions	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec
24	11:28:52	DROP TABLE bank_backup	0 row(s) affected	0.016 sec

SECTION 2 : DML (DATA MANIPULATION LANGUAGE)

DML COMMANDS:

1. INSERT
2. UPDATE
3. DELETE
4. SELECT

STEP 1 : INSERT SAMPLE DATA

```
27 •  INSERT INTO customer_transactions
28    (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
29    VALUES
30    (101, 'Rohini', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),
31    (102, 'Pranitha', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
32    (103, 'Charan', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),
33    (104, 'Manasa', 'Vizag', 'Deposit', 8000, '2024-01-10'),
34    (105, 'Vaishnavi', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
35    (106, 'Rahasyaw', 'Vizag', 'Deposit', 15000, '2024-01-12'),
36    (107, 'Arjun', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
37    (108, 'Haasini', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
38    (109, 'Sneha', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
39    (110, 'Madhu', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
40
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
21	11:25:57	TRUNCATE TABLE customer_transactions	0 row(s) affected	0.015 sec
22	11:26:56	DROP TABLE bank_backup	Error Code: 1051. Unknown table 'bank_db.bank_backup'	0.000 sec
23	11:28:30	CREATE TABLE bank_backup AS SELECT * FROM customer_transactions	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec
24	11:28:52	DROP TABLE bank_backup	0 row(s) affected	0.016 sec
25	11:32:53	INSERT INTO bank_transactions VALUES (101,'Rohini','Hyderabad','Deposit',5000,'...	Error Code: 1146. Table 'bank_db.bank_transactions' doesn't exist	0.000 sec
26	11:33:35	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transa...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec

STEP 2 –

```
41 •  INSERT INTO customer_transactions
42    (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
43    VALUES
44    (111, 'Harshika', 'Vizag', 'Deposit', 7000, '2024-01-18');
45
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
23	11:28:30	CREATE TABLE bank_backup AS SELECT * FROM customer_transactions	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec
24	11:28:52	DROP TABLE bank_backup	0 row(s) affected	0.016 sec
25	11:32:53	INSERT INTO bank_transactions VALUES (101,'Rohini','Hyderabad','Deposit',5000,'...	Error Code: 1146. Table 'bank_db.bank_transactions' doesn't exist	0.000 sec
26	11:33:35	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transa...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec
27	11:35:58	INSERT INTO bank_transactions VALUES (111,'Harshika','Vizag','Deposit',7000,'20...	Error Code: 1146. Table 'bank_db.bank_transactions' doesn't exist	0.000 sec
28	11:36:50	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transa...	1 row(s) affected	0.015 sec

STEP – 3 UPDATE

```
46 •  UPDATE customer_transactions
47    SET amount = 5000
48    WHERE txn_id = 105;
49
50
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
25	11:32:53	INSERT INTO bank_transactions VALUES (101,'Rohini','Hyderabad','Deposit',5000,'...	Error Code: 1146. Table 'bank_db.bank_transactions' doesn't exist	0.000 sec
26	11:33:35	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transa...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec
27	11:35:58	INSERT INTO bank_transactions VALUES (111,'Harshika','Vizag','Deposit',7000,'20...	Error Code: 1146. Table 'bank_db.bank_transactions' doesn't exist	0.000 sec
28	11:36:50	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transa...	1 row(s) affected	0.015 sec
29	11:38:08	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	Error Code: 1146. Table 'bank_db.bank_transactions' doesn't exist	0.000 sec
30	11:38:39	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec

STEP – 4 DELETE

```
50 •  DELETE FROM customer_transactions
51    WHERE txn_id = 111;
52
53
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
27	11:35:58	INSERT INTO bank_transactions VALUES (111,'Harshika','Vizag','Deposit',7000,'20...	Error Code: 1146. Table 'bank_db.bank_transactions' doesn't exist	0.000 sec
28	11:36:50	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transa...	1 row(s) affected	0.015 sec
29	11:38:08	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	Error Code: 1146. Table 'bank_db.bank_transactions' doesn't exist	0.000 sec
30	11:38:39	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
31	11:39:57	DELETE FROM bank_transactions WHERE txn_id = 111	Error Code: 1146. Table 'bank_db.bank_transactions' doesn't exist	0.000 sec
32	11:40:50	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected	0.000 sec

```
75
76 •  CREATE INDEX idx_books_isbn ON Books(isbn);
77 •  SELECT * FROM Books WHERE isbn = '9780451524935';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

book_id	title	isbn	published_year
3	1984	9780451524935	1949
NULL	NULL	NULL	NULL

Result Grid

Books 7 ×

Apply

Output

Action Output

#	Time	Action	Message	Duration / Fetch
27	09:07:38	CREATE INDEX idx_books_isbn ON Books(isbn)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
28	09:07:38	SELECT * FROM Books WHERE isbn = '9780451524935' LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

STEP – 5

```
53 •  SELECT *
54    FROM customer_transactions;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Rohini	Hyderabad	Deposit	5000.00	2024-01-05	NULL
102	Pranitha	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
103	Charan	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Manasa	Vizag	Deposit	8000.00	2024-01-10	NULL
105	Vaishnavi	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL

customer_transactions 1 | Apply | Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
29	11:38:08	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	Error Code: 1146. Table 'bank_db.bank_transactions' doesn't exist	0.000 sec
30	11:38:39	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
31	11:39:57	DELETE FROM bank_transactions WHERE txn_id = 111	Error Code: 1146. Table 'bank_db.bank_transactions' doesn't exist	0.000 sec
32	11:40:50	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected	0.000 sec
33	11:42:24	SELECT * FROM bank_transactions LIMIT 0, 1000	Error Code: 1146. Table 'bank_db.bank_transactions' doesn't exist	0.000 sec
34	11:42:51	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

STEP – 6 SELECT WITH WHERE

```
55    WHERE transaction_type = 'Deposit';
56
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Rohini	Hyderabad	Deposit	5000.00	2024-01-05	NULL
103	Charan	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Manasa	Vizag	Deposit	8000.00	2024-01-10	NULL
106	Rahasyaw	Vizag	Deposit	15000.00	2024-01-12	NULL
108	Haasini	Hyderabad	Deposit	9000.00	2024-01-14	NULL

customer_transactions 2 | Apply | Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
33	11:42:24	SELECT * FROM bank_transactions LIMIT 0, 1000	Error Code: 1146. Table 'bank_db.bank_transactions' doesn't exist	0.000 sec
34	11:42:51	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
35	11:44:27	WHERE transaction_type = 'Deposit'	Error Code: 1064. You have an error in your SQL syntax; check the manual that corr...	0.000 sec
36	11:44:48	WHERE transaction_type = 'Deposit'	Error Code: 1064. You have an error in your SQL syntax; check the manual that corr...	0.000 sec
37	11:45:16	WHERE transaction_type = 'Deposit'	Error Code: 1064. You have an error in your SQL syntax; check the manual that corr...	0.000 sec
38	11:45:24	SELECT * FROM customer_transactions WHERE transaction_type = 'Deposit' LIMI...	6 row(s) returned	0.000 sec / 0.000 sec

STEP 7 SELECT WITH ORDER BY

```
59 •  SELECT *
60    FROM customer_transactions
61    ORDER BY amount DESC;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
106	Rahasyaw	Vizag	Deposit	15000.00	2024-01-12	NULL
103	Charan	Vijayawada	Deposit	12000.00	2024-01-08	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL
108	Haasini	Hyderabad	Deposit	9000.00	2024-01-14	NULL
104	Manasa	Vizag	Deposit	8000.00	2024-01-10	NULL
101	Rohini	Hyderabad	Deposit	5000.00	2024-01-05	NULL
105	Vaishnavi	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL

customer_transactions 3

Output

Action Output

#	Time	Action	Message
40	11:56:08	RENAME TABLE bank_transactions TO customer_transactions	Error Code: 1050. Table 'customer_transactions' already exists
41	11:56:47	SELECT * FROM customer_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned

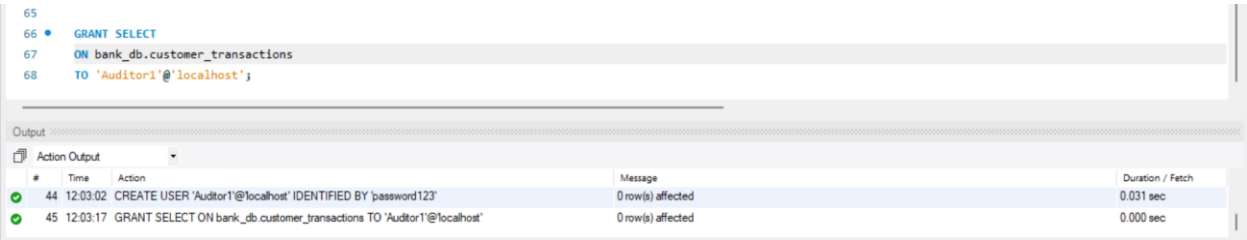
SECTION 3 DCL (DATA CONTROL LANGUAGE) :

DCL COMMANDS:

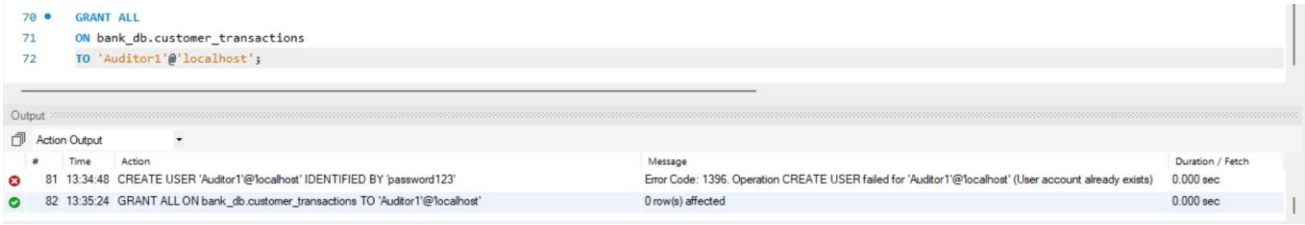
1. GRANT

2. REVOKE

- **GRANT SELECT**



GRANT ALL PRIVILEGES:



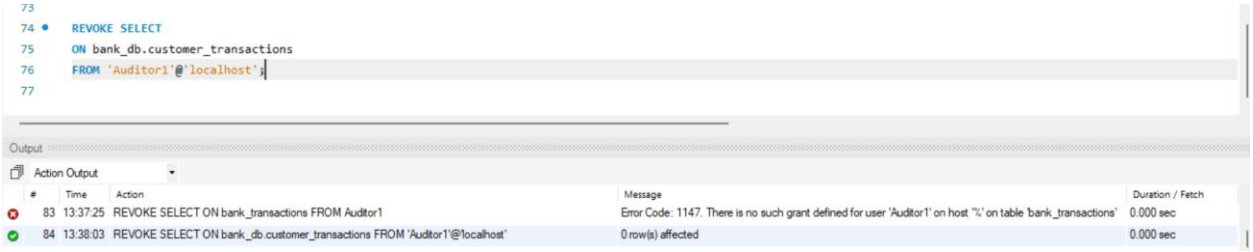
```
70 •  GRANT ALL
71    ON bank_db.customer_transactions
72    TO 'Auditor1'@'localhost';
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
81	13:34:48	CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY 'password123'	Error Code: 1396. Operation CREATE USER failed for 'Auditor1'@'localhost' (User account already exists)	0.000 sec
82	13:35:24	GRANT ALL ON bank_db.customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected	0.000 sec

REVOKE SELECT :



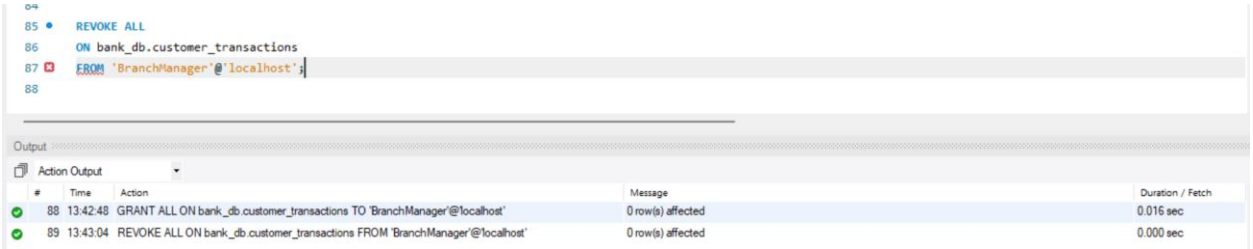
```
73
74 •  REVOKE SELECT
75    ON bank_db.customer_transactions
76    FROM 'Auditor1'@'localhost';
77
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
83	13:37:25	REVOKE SELECT ON bank_transactions FROM Auditor1	Error Code: 1147. There is no such grant defined for user 'Auditor1' on host '%' on table 'bank_transactions'	0.000 sec
84	13:38:03	REVOKE SELECT ON bank_db.customer_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected	0.000 sec

REVOKE ALL :



```
85 •  REVOKE ALL
86    ON bank_db.customer_transactions
87    FROM 'BranchManager'@'localhost';
88
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
88	13:42:48	GRANT ALL ON bank_db.customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected	0.016 sec
89	13:43:04	REVOKE ALL ON bank_db.customer_transactions FROM 'BranchManager'@'localhost'	0 row(s) affected	0.000 sec

SECTION 4 : TCL (TRANSACTION CONTROL LANGUAGE)

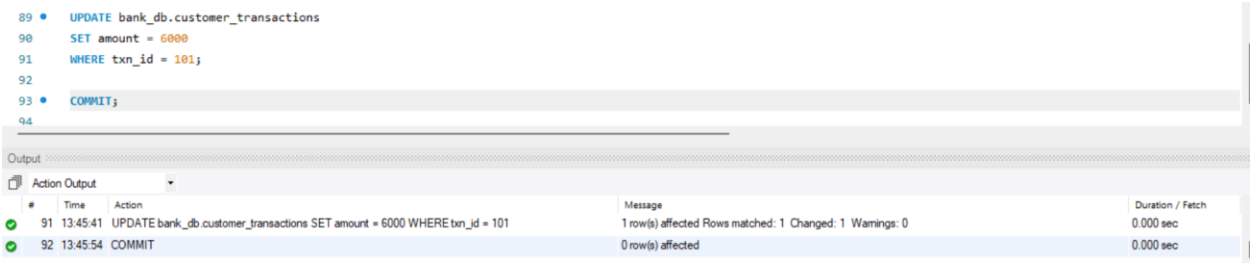
TCL COMMANDS:

1. COMMIT

2. SAVEPOINT

3. ROLLBACK

COMMIT :



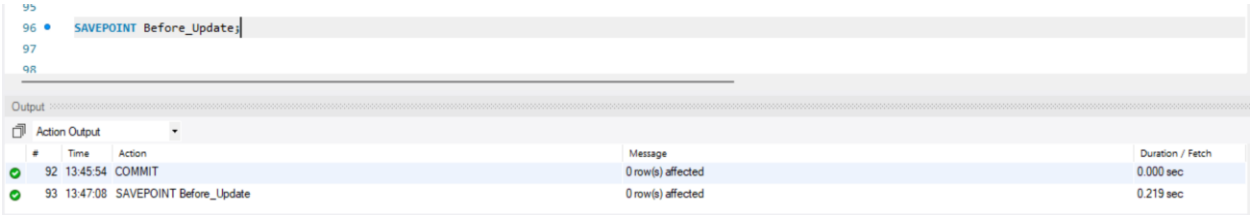
```
89 •  UPDATE bank_db.customer_transactions
90    SET amount = 6000
91    WHERE txn_id = 101;
92
93 •  COMMIT;
94
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
91	13:45:41	UPDATE bank_db.customer_transactions SET amount = 6000 WHERE txn_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
92	13:45:54	COMMIT	0 row(s) affected	0.000 sec

SAVEPOINT :



```
95
96 •  SAVEPOINT Before_Update;
97
98
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
92	13:45:54	COMMIT	0 row(s) affected	0.000 sec
93	13:47:08	SAVEPOINT Before_Update	0 row(s) affected	0.219 sec

ROLLBACK TO SAVEPOINT :

```
99 •  UPDATE bank_db.customer_transactions
100   SET amount = 99999
101   WHERE txn_id = 102;
102
103 •  ROLLBACK TO Before_Update;
104
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
104	13:51:10	UPDATE bank_db.customer_transactions SET amount = 99999 WHERE txn_id = 102	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.000 sec
105	13:51:13	ROLLBACK TO Before_Update	0 row(s) affected	0.000 sec

FULL TCL EXAMPLE :

```
105 •  UPDATE bank_db.customer_transactions
106    SET amount = 7000
107    WHERE txn_id = 101;
108
109 •  SAVEPOINT SP1;
110
111 •  UPDATE bank_db.customer_transactions
112    SET amount = 9000
113    WHERE txn_id = 102;
114
115 •  ROLLBACK TO SP1;
116
117 •  COMMIT;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
107	13:53:06	UPDATE bank_db.customer_transactions SET amount = 7000 WHERE txn_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
108	13:53:10	SAVEPOINT SP1	0 row(s) affected	0.000 sec
109	13:53:14	UPDATE bank_db.customer_transactions SET amount = 9000 WHERE txn_id = 102	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
110	13:53:16	ROLLBACK TO SP1	0 row(s) affected	0.000 sec
111	13:53:19	COMMIT	0 row(s) affected	0.000 sec